

Soc internship day 25

1. Explain UEBA

UEBA (User and Entity Behavior Analytics) is a cybersecurity technology that uses machine learning and analytics to establish a baseline of normal behavior for users and devices (entities) and then detects unusual activities that may indicate security threats.

How UEBA works

1. Collects data from logs, endpoints, network devices, and applications.
2. Learns normal behavior patterns.
3. Detects deviations from the baseline.
4. Assigns risk scores to suspicious activities.
5. Alerts security teams for investigation.

Example

If an employee usually logs in from Kochi between 9 AM and 6 PM, but suddenly logs in at 2 AM from another country and downloads large amounts of data, UEBA may flag this as suspicious

2. Differentiate SIEM and UEBA

Feature	SIEM	UEBA
Full Form	Security Information and Event Management	User and Entity Behavior Analytics
Focus	Log collection and correlation	Behavioral analysis
Detection Method	Rules and signatures	Machine learning and anomaly detection
Threat Detection	Known threats	Known and unknown threats
Insider Threat Detection	Limited	Strong
Alerts	Rule-based	Risk-based and behavior-based
Data Source	Security logs and events	User, device, and application behavior

3. Analyze Insider Threat Scenario

Scenario

An employee in the finance department plans to steal company data before resigning.

Activities

1. Accesses files not normally used.
2. Downloads large amounts of sensitive data.
3. Uses a USB drive to copy files.
4. Logs in outside normal working hours.
5. Uploads data to cloud storage.

How UEBA Detects It

- Identifies unusual file access patterns.
- Detects abnormal data transfer volume.
- Notices unusual login times.
- Correlates multiple suspicious actions.
- Raises the user's risk score.

Response

1. Investigate alerts.
2. Review user activities.
3. Restrict account privileges if necessary.
4. Preserve logs and evidence.
5. Escalate to HR and management if malicious activity is confirmed.

4. Abnormal Behavior Indicators

Common indicators that UEBA can detect:

User Indicators

- Login from unusual locations.
- Login at unusual times.
- Multiple failed login attempts.
- Privilege escalation attempts.
- Access to sensitive files not normally used.

Device Indicators

- New or unknown device access.
- Disabled security software.
- Unusual network connections.
- Large outbound data transfers.

Network Indicators

- Connections to suspicious IP addresses.
- Sudden increase in network traffic.
- Data exfiltration activity.
- Unusual protocol usage.

Application Indicators

- Access to applications not normally used.
- Excessive database queries.
- Large data exports.
- Creation of unauthorized accounts.

Key Point

UEBA is especially effective at detecting insider threats, compromised accounts, and anomalous user behavior that traditional rule-based security tools may miss.